

AgriScout — comparative study of value-based and policy-gradient RL

Task. A rover must keep a 6×9 crop field healthy over a 150-step episode: irrigate stressed cells, spray pest hotspots, and return to a depot to refill, while battery, water and pesticide deplete. Four algorithms — DQN, PPO, A2C and a from-scratch REINFORCE — are tuned and compared under an identical environment-step budget.

Headline result. PPO and DQN finish in a dead heat — **79.4%** and **79.3%** of a privileged scripted oracle's score, both at a **65% success rate** — with A2C at 53.1% and vanilla REINFORCE at 19.2%. The single largest determinant of performance was not the choice of algorithm or its hyperparameters; it was the **shape of the reward function**, which in its first version made the task provably unlearnable for every algorithm tried.

1. Environment

Observation	<code>Box(0, 1, (176,))</code> — health grid, pest grid, irrigation timers, rover position, battery, water, pesticide, step fraction, + 8 egocentric target features
Action	<code>Discrete(9)</code> — <code>MOVE_{N,S,E,W}</code> , <code>SCAN</code> , <code>IRRIGATE</code> , <code>SPRAY</code> , <code>RETURN_TO_DEPOT</code> , <code>WAIT</code>
Horizon	150 steps, 6 × 9 field
Success	mean health ≥ 0.60 and mean pest ≤ 0.15 at the final step

Pests seed as 3–4 localized hotspots that grow and spread to neighbours with a *saturating* probability, so an untreated infestation costs linearly rather than exponentially. Health decays in proportion to local pest severity and recovers when local pest is low — recovery is gated on the *cell's own* pest level, never the field mean, so a cleaned cell always heals and collapse is never absorbing.

Why a cell grid is the right model here, not a simplification

A cell layout can be a lazy abstraction; here it is the domain's own discretization. Precision agriculture genuinely operates on **management zones**: multispectral drone and satellite imagery arrives as a raster, per-cell vegetation and pest-pressure indices are computed from it, and variable-rate applicators are driven by a raster treatment map. The grid *is* the data structure the real problem uses, so this is a faithful model rather than a reduced one. Concretely, the environment differs from a grid world on every axis that matters:

grid world	AgriScout
discrete tile types	two continuous fields in $[0, 1]$ (health, pest) over 54 cells, plus per-cell irrigation timers
static map	coupled stochastic dynamics — pest growth, probabilistic 4-neighbour transmission, pest-driven health decay, recovery
reach a goal tile	multi-objective control : hold mean health ≥ 0.60 <i>and</i> mean pest ≤ 0.15 simultaneously
unconstrained movement	three depleting resources (battery, water, pesticide) with a spatially-located refill depot — a routing/logistics problem
deterministic	fresh layout, hotspot count and hotspot placement every episode ; stochastic spread thereafter

The pest process is a probabilistic cellular model with saturating transmission — structurally an agent-based epidemiological simulation, not a maze. The state space is continuous and effectively unbounded (a 176-dimensional real vector), not enumerable.

The task is not solvable by trivial behaviour, and I measured that rather than assuming it: a uniform-random policy scores 0% success, and so does a do-nothing (`WAIT`) policy — it is *worse* than random (–21.5 vs –13.9), because pests left alone overrun the field. Success requires coordinated multi-step planning: reach hotspots before they spread, treat them, irrigate the worst cells, and return to the depot before the tanks run dry.

Edge cases are first-class, not undefined. Every action is legal in every state, and each degenerate use has an explicit, penalised outcome rather than a silent no-op: spraying an already-clean cell, irrigating an already-healthy cell, acting with an empty tank, driving into a boundary, and calling for the depot while away from it. The demo surfaces these as "wasted" steps, which is how the random policy is visibly distinguished from a trained one (42 wasted actions vs PPO's 0 on the same seed).

Production path. The observation is exactly what a drone imagery pipeline already produces (per-cell health and pest indices + machine telemetry), and the output is a treatment decision per cell. §8 demonstrates the serving half of that pipeline: the trained policies run behind a JSON HTTP API that a farm-management frontend consumes with no RL dependency.

Reward

Reward is **action-attributed**: the agent is paid for health it adds and pest it removes, minus waste penalties (clean-spray, overwater, empty-tank, wall-bump) and a small time cost. Natural pest growth and passive decay change the world but never the reward, so per-step return is not polluted by dynamics the agent cannot control.

On top of that sits **potential-based shaping** (Ng, Harada & Russell, 1999):

```
Φ(s) = 25·mean_health - 20·mean_pest + 1.0·(-distance to nearest work)
r += Φ(s') - Φ(s)
```

Shaping uses $\gamma = 1.0$, so the total it can pay over an episode telescopes to exactly $\Phi(s_T) - \Phi(s_0)$. Two consequences matter: the agent is paid precisely for the *net* field improvement it causes, and any cycle in Φ sums to exactly zero, so the term cannot be farmed. Coefficients were set from measured trajectory spread, not intuition — oracle and random policies differ by +0.487 mean health and -0.570 mean pest.

2. Method

- **Budget in environment steps**, so algorithms with different update mechanics are compared fairly: 100k per sweep configuration (250k for REINFORCE, justified in §5), 400k for every final.
- **Two-stage protocol**. 40 sweep configurations (10 per algorithm) rank hyperparameters within an algorithm; the best configuration per algorithm is then retrained from scratch at 400k steps for the cross-algorithm comparison.
- **Three disjoint seed sets**. Tuning and finals are scored on seeds 9000–9019; generalization additionally uses 10000–10049, which are touched at no point during training or model selection.
- **Two reference policies** bound the scale: a uniform-random policy (–13.88) and a privileged scripted oracle (+14.55). Results are reported as *% of oracle*, which is the only figure comparable across reward-function versions.
- **Both action-selection modes** (greedy argmax and sampled) are measured at every stage — §6 explains why this turned out to be essential rather than cosmetic.

3. The central finding: winnable ≠ learnable

The first reward function (`archive/v0_unlearnable/`) passed a scripted-oracle winnability gate — the oracle scored +17.03 with a 100% success rate — and was nevertheless impossible to learn. Across 40 configurations and four 400k finals, about **8M environment steps, not one run recorded a single success**, and three of four algorithms finished *worse than random*.

Decomposing the oracle's return into its dense (per-step) and terminal parts showed why:

policy	total	dense	terminal bonus
oracle	+17.03	–2.97	+20.00
random	–5.13	–5.13	0

The entire *learnable* gap between optimal and random behaviour was **2.16 reward over 150 steps — 0.014 per step** — against per-episode noise of $\sigma \approx 2\text{--}5$. Signal-to-noise below 1. Ninety percent of the oracle's advantage sat in a single all-or-nothing terminal bonus, gated on two simultaneous thresholds, that no agent ever once received and

therefore could never learn from.

A second defect compounded it: the navigation shaping had been scaled against the per-*episode* time cost instead of the per-*step* one, so moving one step toward work scored **-0.015**. The term penalised precisely the behaviour it was added to encourage.

	v0	v1
dense gap (oracle – random)	2.16	23.43
terminal bonus as % of total gap	90%	18%
net reward per step toward work	-0.015	+0.057
best agent	24.6% of oracle, 5% success	79.4% of oracle, 65% success

A scripted-oracle gate proves a task is *winnable*. It says nothing about whether it is *learnable*. Those are different properties, and only the second predicts whether training will work. Two guards now enforce the distinction: `tests/test_oracle.py` asserts a minimum dense-reward gap and that approaching work is net-positive, and `scripts/learnability_gate.py` trains a real agent briefly and requires it to beat random *before any sweep compute is spent*.

A second representational fix

With the reward corrected, PPO began learning but stalled, oscillating between `MOVE_E` (0.22) and `MOVE_W` (0.26) — driving back and forth without progress. The cause was representational: the grid is flattened into the observation vector, so an MLP policy had to internally learn an argmax over 54 cells *and* a Manhattan distance purely from unstructured floats, before it could act on either.

Eight **egocentric target features** (relative offset to the nearest hotspot and to the lowest-health cell, plus local health/pest) state the navigation target directly. They are a pure function of already-observable state — the scripted oracle computes exactly the same argmax — so this adds no privileged information; it removes a representation burden unrelated to the control problem. PPO's improvement over random went from +4.52 to **+21.75** on the same budget.

4. Hyperparameter tuning

Full tables: `assets/tables/hparams_{dqn,ppo,a2c,reinforce}. {csv,md}`. Sensitivity plot: `assets/figures/fig4_hparam_sensitivity.png`.

△ **Methodological caveat.** Each configuration uses a *different random seed*, so a single-run difference between two configurations conflates the hyperparameter effect with seed noise. Where a factor is claimed to matter below, the claim is supported by aggregating over the several configurations that share that factor level, not by any one run. Isolating hyperparameter effects properly would need multiple seeds per configuration — see §8.

DQN — `lr` ∈ {1e-3, 5e-4, 3e-4}, `γ` ∈ {0.98, 0.99}, `buffer_size` ∈ {50k, 200k}

Discount factor dominates: $\gamma = 0.98$ averages -4.41 against $\gamma = 0.99$'s -10.79, and $\gamma = 0.98$ takes all three top places. The worst runs pair $\gamma = 0.99$ with the largest learning rate — a long effective horizon plus an aggressive step size, the classic recipe for Q-value divergence in a bootstrapped learner.

Replay buffer size matters, and smaller is better here: 50k averages -5.20 against 200k's -8.72, and every one of the top two runs uses the small buffer. This is the opposite of the usual "bigger replay is better" intuition, and the reason is the budget: at 100k sweep steps a 200k-transition buffer never even fills, so it keeps serving transitions collected during the early near-random phase. The small buffer cycles roughly twice per run, keeping the sampled distribution closer to the current policy. Replay size is not a free parameter — it interacts with the training budget.

Buffer size replaced `exploration_fraction` in the grid after an earlier sweep measured the latter as having no effect (0.1 → -5.75 vs 0.2 → -5.61); with a densely shaped reward, the ϵ schedule is not what limits this agent.

The best configuration (`dqn_04`: `lr` 5e-4, γ 0.98, 50k buffer) retrained at 400k steps produced the study's strongest final agent on field health (0.858) and matched PPO on reward.

PPO — $\text{lr} \in \{3\text{e-}4, 1\text{e-}3\}$, $\text{n_steps} \in \{256, 512\}$, $\text{ent_coef} \in \{0, 0.01, 0.05\}$

Entropy regularisation is the clearest single factor in the whole study, and it is non-monotonic:

ent_coef	mean stochastic reward	best
0.00	1.63	4.53
0.01	3.71	5.54
0.05	1.98	2.25

Too little entropy and the policy commits early to a mediocre routine; too much and it never commits at all, dithering permanently. 0.01 is the balance point, and the best overall configuration (`ppo_03`) pairs it with the longer 512-step rollout.

A2C — $\text{lr} \in \{7\text{e-}4, 3\text{e-}4\}$, $\text{n_steps} \in \{5, 16\}$, $\text{ent_coef} \in \{0, 0.01, 0.05\}$

A2C was the least stable learner. Rollout length is the factor that separates its runs: $\text{n_steps} = 5$ **averages -8.01 against $\text{n_steps} = 16$'s -12.64** , taking four of the top five places, with three of the four long-rollout runs in the bottom half. With no replay buffer and a short-horizon advantage estimate, A2C depends on frequent updates. Its learning curve (fig. 1) shows the largest late-training oscillations of any algorithm, including a sharp collapse and recovery near 370k steps.

REINFORCE — $\text{lr} \in \{1\text{e-}3, 5\text{e-}4\}$, $\text{baseline} \in \{\text{none}, \text{mean}, \text{value}\}$, $\text{episodes_per_batch} \in \{2, 4\}$

The baseline comparison is the cleanest result in the study, and it reproduces the textbook variance-reduction ordering exactly:

baseline	mean stochastic reward	best	n
none (raw returns)	-18.97	-18.49	2
mean (batch-mean subtracted)	-12.49	-11.07	4
value (learned critic)	-9.40	-6.39	4

All four top runs use the learned critic; both `none` runs are last. To keep this comparison meaningful, advantage rescaling divides by the standard deviation but **never re-centres** — otherwise `none` and `mean` become mathematically identical and the experiment measures nothing.

Three implementation fixes were required before REINFORCE learned at all. Updating from a *single* episode left the policy at entropy 1.95 of a 2.20 maximum after 150k steps — still essentially uniform. The critic, fitted jointly at the policy's learning rate, never dropped below a loss of ~15, so `returns - V` was still effectively raw returns. And without gradient clipping, every learning rate $\geq 3\text{e-}3$ drove entropy to 0 within a few updates and froze the policy on a single action (a measured, degenerate always-`WAIT` run scoring -21.52). Batching, a separately-optimised critic, and gradient clipping together took it from "no learning" to a steadily climbing curve.

5. Results

Reference policies on seeds 9000–9019: random **-13.88**, oracle **+14.55**.

algo	det	stochastic	success	final health	held-out	gen. gap	% of oracle
PPO	1.73	8.68	0.65	0.794	7.09	+1.83	79.4%
DQN	8.57	8.66	0.65	0.858	7.10	+2.32	79.3%
A2C	-4.65	1.20	0.45	0.722	-0.01	+4.26	53.1%
REINFORCE	-21.65	-8.42	0.00	0.489	-7.06	+1.48	19.2%

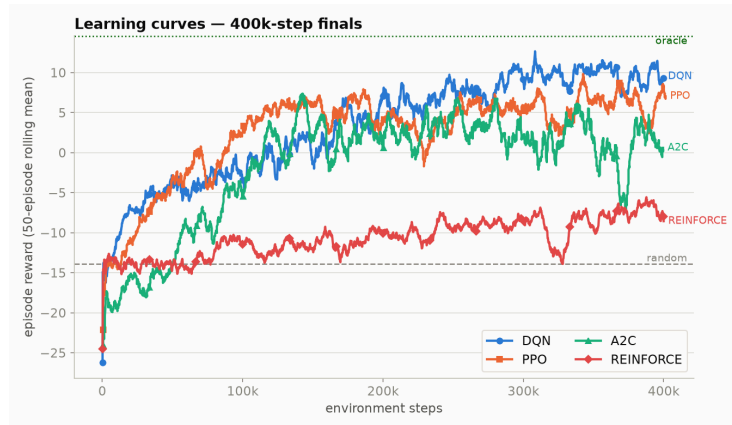


Fig 1 — 50-episode rolling mean of episode reward, all four methods overlaid, against the random and oracle reference lines.

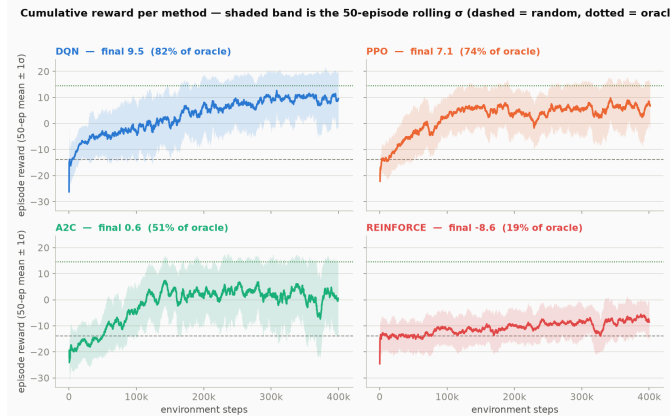


Fig 6 — the same curves per method with a $\pm 1\sigma$ band. The overlay ranks methods; this panel shows each one's stability, which the overlay hides. A2C's band is visibly the widest.

PPO and DQN finish level (79.4% vs 79.3%); the policy-gradient family is not uniformly better or worse than the value-based one. The meaningful split is not value-based vs policy-gradient but *how much a method reuses its experience*. PPO (clipped multi-epoch updates over a rollout buffer) and DQN (a replay buffer) both extract many gradient updates per environment step. A2C, which discards each rollout after one update, sits well behind them, and vanilla REINFORCE — one update per batch of complete episodes — is last by a wide margin.

That ordering shows up directly in the learning curves: DQN and PPO rise fastest and plateau around 250k steps; A2C climbs more slowly and oscillates hard; REINFORCE is still climbing at 400k, where the others have long since flattened.

On REINFORCE's budget. Per 150k environment steps, PPO performs roughly 23,000 gradient updates (256 steps $\times$ 4 envs, 10 epochs, minibatch 64); vanilla REINFORCE performs about 500. It fails a 150k-step learnability gate and passes at 400k, which is why its sweep budget was raised to 250k — at 100k its runs are indistinguishable noise and ranking them would measure the seed rather than the hyperparameters. Its weakness here is a real property of Monte-Carlo policy gradients, not a defect to be tuned away.

Learning dynamics — the objective and entropy curves

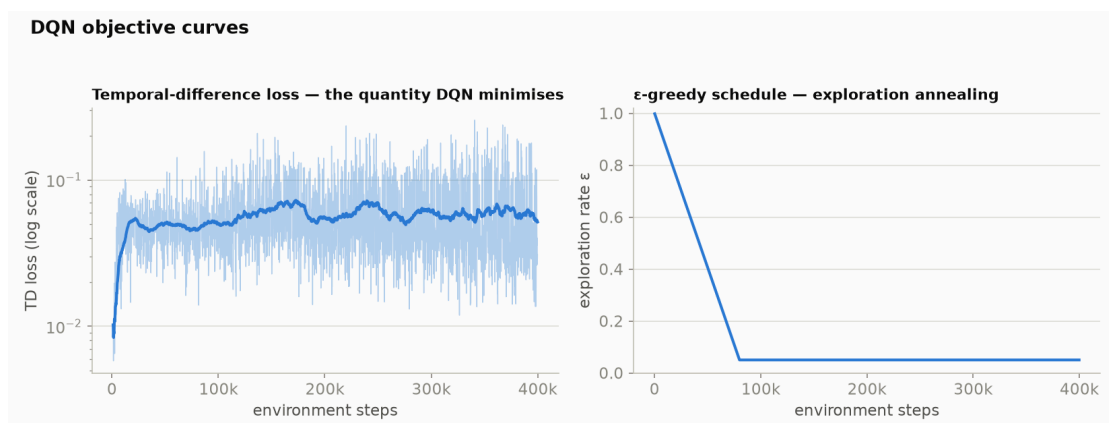


Fig 7 — DQN's temporal-difference loss (log scale) and its ϵ -greedy schedule.

DQN's TD loss does **not** decrease over training — it rises from ~0.01 and settles around 0.05–0.07. That is expected rather than alarming for a bootstrapped learner and is worth stating explicitly, because a falling loss is the wrong thing to look for here: the regression target $r + \gamma \cdot \max Q(s')$ is itself moving as the policy improves, and as ϵ anneals the agent reaches richer, higher-reward states it had never visited while exploring. Loss magnitude tracks target variance, not policy quality — reward (fig 1) is the quantity that improves.

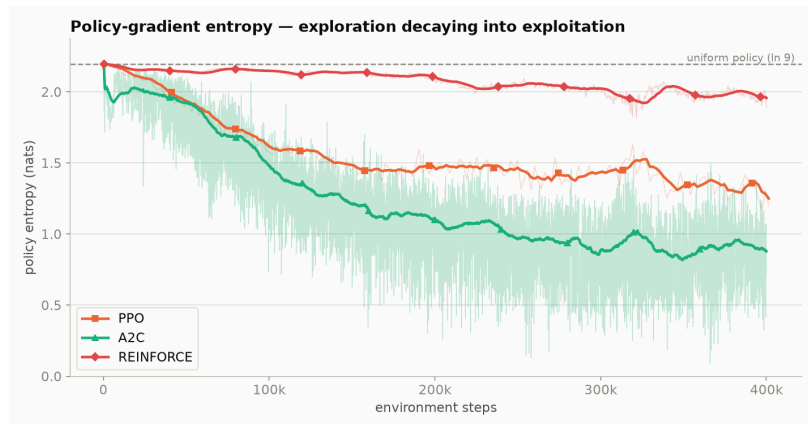


Fig 8 — policy entropy for the three policy-gradient methods, against the $\ln 9$ uniform-policy ceiling.

Entropy is the clearest single diagnostic in the study, and it explains the final ranking almost by itself:

method	entropy 0 → 400k	reading
PPO	2.20 → ~1.25	steady, controlled commitment — the healthiest trace
A2C	2.20 → ~0.90	commits hardest and earliest; matches its wide reward band and late collapses
REINFORCE	2.20 → ~1.96	barely leaves the uniform policy

REINFORCE ends 400k steps still at 89% of maximum entropy. It is not choosing badly so much as *barely choosing at all*

— which is exactly what ~500 gradient updates buys, and is the direct cause of its 0% success rate. This is also why its greedy policy is so much worse than its sampled one (§6): an argmax over a near-uniform distribution is close to arbitrary.

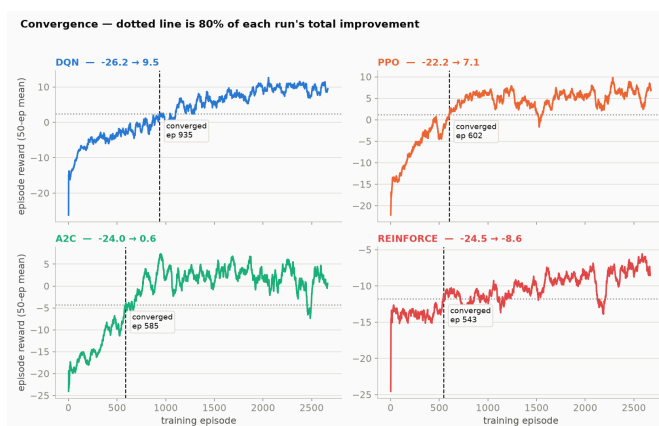


Fig 9 — convergence, defined as the first episode whose rolling mean reaches 80% of the run's total improvement (start → final).

Convergence episodes span 543–935, but the metric measures

when each run captured most of its own gain

, not how much gain there was — so it must never be read alone. REINFORCE "converges" earliest (episode 543) precisely because it has the least improvement to capture, while DQN is latest (935) because it kept improving longest and ends highest on field health. Read fig 9 together with the final values in the panel titles.

6. Deterministic vs stochastic action selection

The three policy-gradient methods are markedly better *sampled* than *greedy*, and the effect is large: across the PPO sweep the mean stochastic-minus-deterministic gap is **+11.59 reward**. The best PPO configuration scores -10.63 greedy and $+5.54$ sampled *on identical weights*.

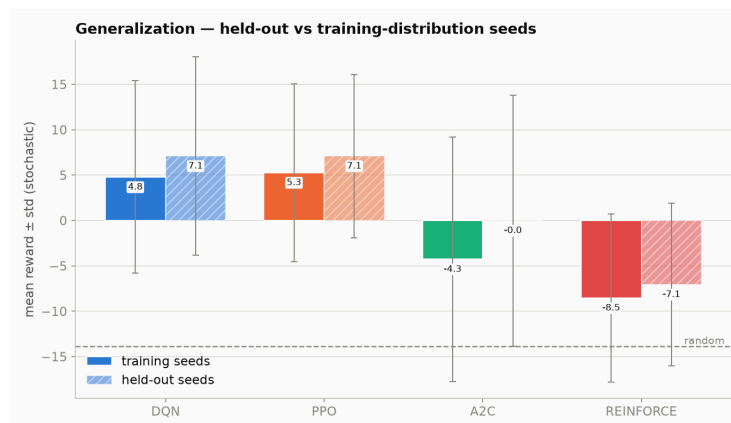
The final DQN agent is the informative exception — 8.57 greedy vs 8.66 sampled, a gap of 0.09 . This is what the split should look like: DQN learns an action-*value* function, so its argmax is over calibrated Q-estimates and is meaningful, whereas a policy-gradient method optimises the sampled distribution directly and its argmax is merely a by-product. That DQN alone is mode-insensitive is evidence the gap is a property of the *policy parameterisation*, not a quirk of this environment.

This is not a curiosity; it changed the experiment twice.

1. **Model selection.** Ranking sweep runs on deterministic reward alone selected a different configuration for *all four* algorithms — in PPO's case discarding the eventual best run because its argmax happens to be poor. Selection now ranks on the better of the two modes.
2. **Reporting.** An early generalization table, scored greedily, showed PPO at -4.18 on held-out seeds and read as a generalization failure. Its stochastic held-out score is **+7.09** — *higher* than its training-distribution score.

The likely mechanism is that the field has many near-equivalent next actions (several cells equally worth treating). A near-tie under argmax resolves to one fixed choice that may be systematically poor, while sampling preserves the diversity the task rewards.

7. Generalization



Every agent scores **higher on the 50 never-seen held-out seeds than on training-distribution seeds** (generalization gap positive for all four: $+1.83$, $+2.33$, $+4.26$, $+1.48$). There is no measurable overfitting to the training seed distribution.

This is expected rather than surprising: every episode draws a fresh field layout, hotspot count and hotspot placement, so an agent never sees the same field twice and there is no fixed layout to memorise. The positive sign is best read as sampling noise between two seed sets of different size (50 vs 20) rather than as genuine better-than-training performance — the error bars in fig. 3 are wide enough to span it.

8. Deployment path — JSON API and web frontend

The trained policies are not left as `.zip` files. `api.py` (FastAPI) exposes them over HTTP as JSON, in the two shapes a product actually needs:

endpoint	purpose
POST /episode	run a whole episode, return every frame — enough for a client to replay it
POST /session → /session/{id}/step	stateful session; the agent picks each action
POST /session/{id}/act	same session, but a client-chosen action — lets a UI drive manually or override
GET /agents	registry with each agent's eval metrics and hyperparameters

Every response is plain JSON — grids as nested arrays, rover pose, resource levels, reward, and whether the success criteria are currently met — so a web or mobile client renders it with no RL dependency at all. The bundled viewer (`assets/demo/index.html`, `three.js`) is served from `/` as a worked example, and the interactive OpenAPI schema is at `/docs`.

```
uv run uvicorn api:app --reload
curl -X POST localhost:8000/episode -H 'Content-Type: application/json' \
  -d '{"agent": "ppo", "seed": 9003, "include_grids": false}'
```

Verified: the `/episode` response for `ppo/seed 9003` reproduces the recorded episode exactly (+15.776 reward, success), and an invalid action is rejected with HTTP 422.

9. Limitations

- **One seed per configuration.** Sweep configurations vary the seed alongside the hyperparameters, so individual run comparisons conflate the two. The aggregate claims in §4 hold across several runs each, but a rigorous sensitivity analysis needs n seeds per configuration — roughly 3× the compute for the sweep stage.
- **Finals are single-run.** Each final is one 400k training run per algorithm, so the 0.1-point gap between PPO (79.4%) and DQN (79.3%) is far inside seed variance and must not be read as a ranking — the honest statement is that they are level. The gap to A2C and REINFORCE is large enough to be safe.
- **A gap to the oracle remains.** The best agent reaches 79.4%; the oracle uses privileged state (exact argmax over the pest grid) and near-perfect resource planning. Closing the rest would likely need longer training or a spatially structured policy (see below).
- **REINFORCE never succeeds** (0% success rate). Its curve is still rising at 400k, so the honest statement is that it is under-trained at this budget, not that it cannot learn the task.

10. Future work

- **Spatial policy architecture.** The grid is flattened for an MLP; a small CNN over the (3, R, C) tensor would preserve adjacency and likely remove the need for the hand-engineered egocentric features entirely.
- **Multi-seed sweeps** to separate hyperparameter effects from seed noise (§9).
- **Drop the dead action.** `SCAN` is a no-op that costs battery and is used by the oracle 0% of the time; removing it shrinks the exploration space by a ninth.
- **Curriculum on hotspot count**, starting easy, to help REINFORCE reach a first success sooner within its very limited update budget.

Appendix — reproducing

```
uv run pytest -q                                # env + learnability gates
uv run python scripts/learnability_gate.py --all  # must pass before spending compute
AGRISCOUT_RESULTS=./logs uv run python -m training.run_all --phase all
AGRISCOUT_RESULTS=./logs uv run python analysis.py
```

Runs are idempotent and resumable (checkpoints + `DONE.json`); the full pipeline takes roughly 30 minutes on 8 physical cores. The interactive episode viewer is `assets/demo/index.html` — see `README.md`.